

Answer **all** questions.

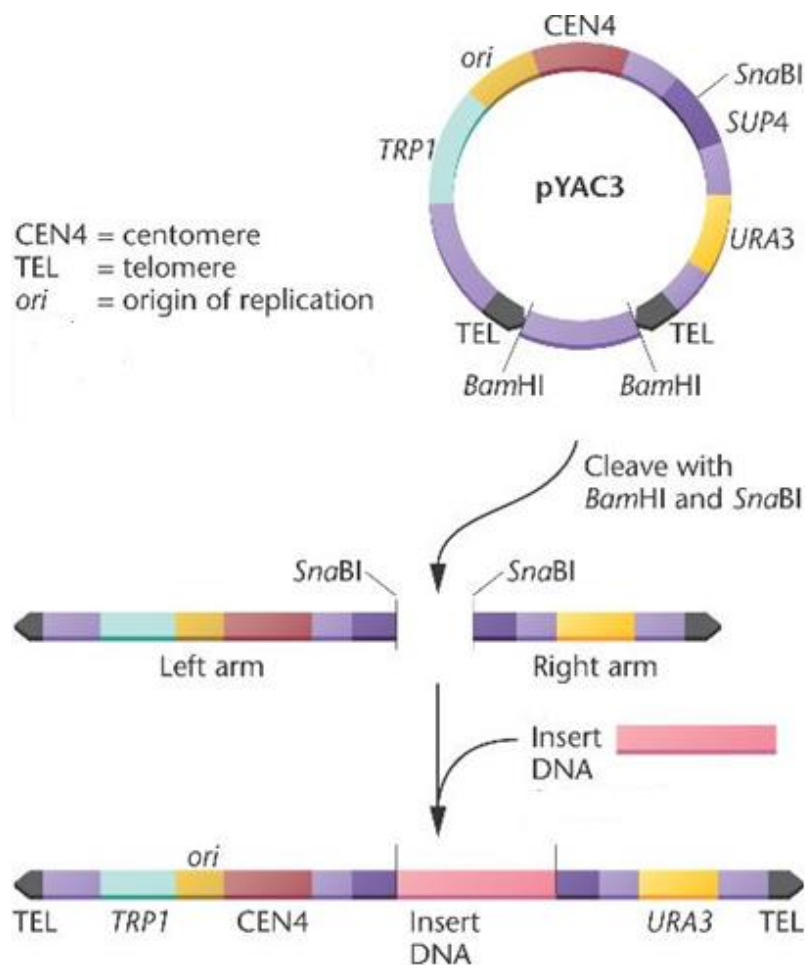
- 1 Yeast is a single-cell eukaryotic organism that can be manipulated and grown in the laboratory much like bacteria. Yeast artificial chromosome (YAC) vectors are designed to act like eukaryotic chromosomes.

pYAC3 contains the following components:

1. an autonomously replicating sequence to function as an origin of replication (*ori*),
2. centromere elements (CEN4) for accurate chromosome segregation to daughter cells during cell division,
3. telomeric sequences (TEL) in order for the ends to be replicated correctly and which also prevent the chromosome from being nibbled away by exonucleases,
4. growth markers (*URA3* and *TRP1*) to select positively for chromosome maintenance, and
5. the *SUP4* gene which encodes for the *SUP4-o* suppressor protein.

The *ADE2* yeast gene encodes an enzyme required in the biosynthetic pathway of adenine. *ADE2* mutants produce a red pigment. When pYAC3 is transformed into a *ADE2* mutant yeast cell, the *SUP4-o* protein suppresses the *ADE2* mutation and white colonies are produced.

Fig. 1.1 shows the pYAC3 vector and the cloning strategy used with pYAC3.



**Fig. 1.1**

(a) With reference to the preamble and Fig. 1.1,

(i) explain how recombinant *ADE2* mutant yeast cells can be selected. [3]

- Grow the yeast in a suitable nutrient medium;
- The **non-recombinant mutant yeast cells** will form **white colonies**;
- This is because the **sup4 gene is intact** and encodes an active *SUP4-o* suppressor protein that suppresses *ADE2* mutation.
- The **recombinant mutant yeast cells** will form **red colonies**.
- This is because the *SUP4* gene has been **insertionally inactivated** when foreign DNA is introduced into the *SnaBI* restriction site;
- Non-functional *SUP4-o* suppressor protein / no *SUP4-o* suppressor protein is synthesised. Hence, no suppression of *ADE2* mutation;

Several important mammalian genes such as the human cystic fibrosis gene, which is 250kb in length, have been cloned using the pYAC3 vector.

Using this information and your own knowledge of cloning eukaryotic genes,

(ii) suggest why pYAC3 is a preferred cloning vector for eukaryotic genes compared to bacterial plasmids. [2]

- To clone **the complete sequences of large genes or gene complexes that exceed the size limit for cloning in bacterial cloning vectors** like plasmids (up to 10 kb);
- Many **sequences** that are unstable, under-represented, or absent when cloned into prokaryotic systems, **remain stable and intact** in YAC clones;
- Important in the **production of gene libraries**;
- Ability to clone large pieces of DNA for use in **large-scale DNA sequencing projects**;

[max 2]

Fig. 1.2 shows the use of a restriction enzyme EcoRV in a separate cloning experiment.

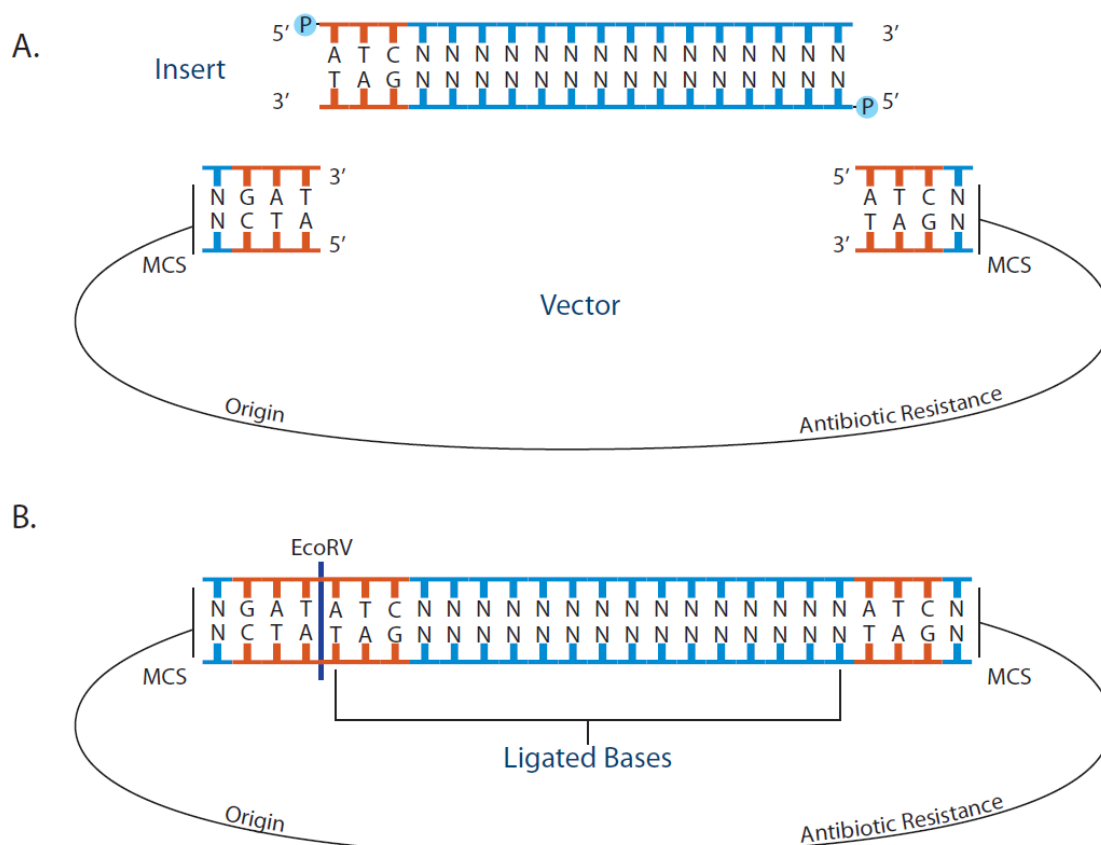


Fig. 1.2

(b) Using the information in Fig. 1.2, explain how a foreign gene can be cloned into the bacterial plasmid shown. [3]

- Flanking regions of foreign gene / insert and plasmid **cleaved with same restriction enzyme, EcoRV**, to create blunt ends; *(must mention in answer to gain full credit)*;
- Double-stranded **linkers** containing specific restriction sites that will generate **sticky ends / staggered ends** when cleaved with suitable restriction enzymes are added;
- The resulting linkers with sticky ends / staggered ends are added to both the linearised plasmid and foreign gene / insert using **terminal transferase** to attach to the blunt ends creating **complementary overhangs** so that annealing can occur via complementary base pairing / formation of hydrogen bonds with each other;
- Add **DNA ligase** to form **phosphodiester bonds** to join the sugar-phosphate backbone to form recombinant DNA molecule;

[max 3]

Transformation is then carried out using competent *E. coli*.

- (c) Outline the additional steps required if the objective of cloning is to harvest the protein products. [3]

**Reject: references to steps describing the process of transformation and / preparation of competent *E. coli* cells.**

1. **Identify the presence of a foreign DNA insert in transformed *E. coli* cell**
2. **by replica plating and selection of the cells that survive on the master plate but not on the replica plate containing antibiotic;**
3. This is followed by **gene probing by DNA hybridization** to identify the recombinant *E. coli* / vectors containing the gene of interest;
4. **Grow the selected host cells in large quantities in a suitable nutrient medium;**
5. These bacteria can then be **induced to produce large amounts of the protein product;**
6. The protein product produced can be **extracted, purified and packaged.**

**[Max 1m for points 4 to 6]**

[Total: 11]

- 2 Osteogenesis imperfecta (OI) is a heterogeneous group of genetic disorders that affect connective tissue integrity. The symptoms of OI include fragile bones, osteoporosis, easy bruising and loose joints. The cause of the disease is linked to mutations in the genes that encode polypeptide chains that comprise type I collagen molecules, the major structural protein of bone.

Table 2.1 shows the mode of inheritance and clinical characteristics of type I and type II OI.

**Table 2.1**

| Type | Mode of inheritance | Clinical characteristics                                                                     |
|------|---------------------|----------------------------------------------------------------------------------------------|
| I    | Autosomal dominant  | Osteoporosis, mild skeletal deformities, mild short stature, easy bruising                   |
| II   | Autosomal dominant  | Lethal form in embryonic development, severe skeletal deformities, beaded ribs and fractures |

(a) With reference to Table 2.1,

- (i) explain the cause of the differences in clinical characteristics between type I and type II, in terms of the different types of gene mutation. [2]
- **Insertion or deletion mutation** leading to **frameshift mutation** / shift in reading frame of gene sequence, resulting in **no production of functional collagen**, which is highlighted by the **severe consequences of type II**;
  - **Substitution or inversion**, changes in parts of the protein sequences resulting in **less functional collagen production**, as indicated by **mild consequences**;
- (ii) explain why conventional gene therapy treatment is not suitable for treating OI. [2]
- Disease is **autosomal dominant**, **normal allele** is **recessive** to the disease allele;
  - inserting the therapeutic gene would not be able to mask the disease phenotype or vice versa;

The cell therapy approach aims to overcome the problems associated with dominant OI mutations by supplying cells that would give rise to osteoblasts, the bone-forming cells, to replace the cells that synthesize abnormal collagen chains.

Osteoblasts are terminally differentiated, non-proliferating cells with a limited lifespan. Osteoblasts arise from mesenchymal stem cells that reside in the bone marrow whereas the osteoclasts, the cells that are involved in bone resorption, are derived from haematopoietic lineage. The ability for bone to regenerate is attributed to stem cells in bone that undergo proliferation and differentiation.

The rationale for cell therapy is that mutant osteoblasts can be replaced by normal osteoblasts derived from the transplanted progenitors. In addition, since the transplanted cells are stem cells, they will self-renew and thus provide treatment for life.

(b) Suggest the meaning of haematopoietic lineage. [1]

- Haematopoietic stem cells that share the similar differentiation pathways;  
**Or**  
Differentiated cells derived from similar stem cells types;

(c) Explain how mesenchymal stem cells may increase bone growth in children with OI. [2]

- Stem cells can **divide by mitosis**;
- and in the presence of the **appropriate signals** can also **differentiate** into **new bone cells**;  
This thus increases the number and type of bone cells present which contribute towards bone growth.

Gene therapy was found to be more effective when used in tandem with anti-sense RNA gene silencing technique. The approach involves the use of double-stranded RNA (dsRNA) sometimes referred to as RNA interference (RNAi). The steps are as follows:

1. Bone marrow cells are taken from diseased individuals.
2. Suppression of diseased RNA expression using RNAi.
3. Transfection of therapeutic gene using a retroviral vector.
4. Injection of treated bone marrow cells back into diseased individuals.

(d) (i) Explain the choice of vector used. [2]

- Retroviral vector is able to target **actively dividing cells, stem cells**;
- Ability to **integrate** the therapeutic gene into the genome for **permanent expression**;

(ii) State the molecular target of gene silencing. [1]

- mRNA;

(iii) Suggest how dsRNA is used to suppress diseased RNA expression. [2]

- dsRNA **denatured to ssRNA** which **binds by complementary base-pairing to mRNA**;
- **results in formation of dsRNA / duplex RNA** which will be recognised by **RNA-degrading enzymes**, thus **no functional mRNA will be available for translation** into non-functional polypeptide;

(iv) Suggest why the *ex vivo* approach was adopted for gene therapy treatment of OI. [2]

- Allows specific **identification of target cells** with therapeutic gene;
- Allows **monitoring of gene expression before reintroduction** as success rate is very low;
- Safer because can **monitor for cancer cells** caused by the **random integration of DNA / insertional mutagenesis by retrovirus** which can knock out tumour suppressor genes or activate proto-oncogenes;

[max 2]

[Total: 14]

- 3 In 2012, permission was granted for a field trial in the UK of genetically modified wheat, *Triticum aestivum*. The wheat carries a gene, taken from peppermint plants, which results in the wheat leaves releasing a volatile, non-toxic chemical, (E)- $\beta$ -farnesene (E $\beta$ f), into the atmosphere.

E $\beta$ f is not only produced by various species of plants. It is also secreted by aphids when they are disturbed by a predator.

Two experiments have been performed into the effect of E $\beta$ f on the behaviour of aphids feeding on leaves in closed containers.

#### Experiment 1

Either 10 cm<sup>3</sup> of air from a syringe that contained plant leaves that secrete E $\beta$ f  
or 10 cm<sup>3</sup> of air from a syringe with no such leaves  
was added to the containers of feeding aphids.

#### Experiment 2

Either 20 cm<sup>3</sup> of air containing 50ng of E $\beta$ f  
or 20cm<sup>3</sup> of air containing no E $\beta$ f  
was added to the containers of feeding aphids.

In both experiments, the number of aphids that stopped feeding and moved away from the food leaves was counted. The results are shown in Table 3.1.

**Table 3.1**

| Air added to containers of feeding aphids                                 | Experiment 1                                                                      |                                                                                       | Experiment 2                                       |                                                  |
|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------|--------------------------------------------------|
|                                                                           | 10 cm <sup>3</sup> air that had been in contact with leaves secreting E $\beta$ f | 10 cm <sup>3</sup> air that had not been in contact with leaves secreting E $\beta$ f | 20 cm <sup>3</sup> air containing 50ng E $\beta$ f | 20 cm <sup>3</sup> air containing no E $\beta$ f |
| Number of aphids in container                                             | 99                                                                                | 113                                                                                   | 132                                                | 106                                              |
| Number of aphids that stopped feeding and moved away from the food leaves | 54                                                                                | 1                                                                                     | 111                                                | 0                                                |

- (a) (i) With reference to Table 3.1, discuss the extent to which the results of these experiments support the idea that E $\beta$ f is an alarm signal for aphids. [3]

- In the presence of E $\beta$ f, large number of aphids stopped feeding and moved away from food source (leaves): 55% OR 54 out of 99 aphids in experiment 1 versus 84% OR 111 out of 132 aphids, in experiment 2 respectively;

**AND**

- In the absence of E $\beta$ f, few / no aphids stopped feeding and moved away from food source: 1 aphid and none, in experiment 1 and 2 respectively;

**OR**

- In experiment 1, 55% OR 54 out of 99 aphids that were exposed to air that had been in contact with leaves secreting E $\beta$ f, stopped feeding and moved away from food source, versus 0.9% OR 1 out of 113 aphids that were exposed to air that had not been in contact with leaves secreting E $\beta$ f, stopped feeding and moved away from food source;

**AND**

- In experiment 2, 84% OR 111 out of 132 aphids that were exposed to air containing 50ng E $\beta$ f, stopped feeding and moved away from food source, versus 0% OR none of the 106 aphids that were exposed to 20 cm<sup>3</sup> air containing no E $\beta$ f, stopped feeding and moved away from food source;

**Max 1m from the following points:**

- The air in Experiment 1 may contain other chemicals / is not pure E $\beta$ f whereas the air in Experiment 2 contains only E $\beta$ f;
- E $\beta$ f concentration in Experiment 2 may be unnaturally high or E $\beta$ f concentration is unknown in Experiment 1;
- However, different volumes of air was used in Experiment 1 (10cm<sup>3</sup>) and Experiment 2 (20 cm<sup>3</sup>);

**[max 3]**

- (ii) Other experiments show that E $\beta$ f attracts predators of aphids, such as ladybirds.

Explain how growing genetically modified wheat secreting E $\beta$ f could increase the yield of wheat. [3]

- Secreted E $\beta$ f stops aphids from settling on wheat plants as shown in the data from Table 3.1;
- E $\beta$ f attracts predators of aphids / ladybirds which will prey on aphids;
- attacked aphids / aphids disturbed by predators / ladybirds will secrete even more E $\beta$ f, which attracts more predators / ladybirds;
- Either way, aphids are not able to feed on wheat / take nutrients from GM wheat;

**[max 3]**

- (iii) Suggest why growing this genetically modified wheat might be acceptable to people who object to the growth of genetically modified insect-resistant maize or cotton. [3]

- gene encoding for E $\beta$ f is already found in peppermint / various plant species' genomes;
- E $\beta$ f is not toxic / harmful to human health;

OR

no new chemical is added to human diet;

- Unlike Bt maize or Bt cotton, E $\beta$ f does not kill insects (since it is naturally secreted from aphids);
- Aphids are thus still available for their predators / minimal impact on the food chain;

[max 3]

In a separate experiment, *Agrobacterium*-mediated transformation was performed on *T. triticum* to produce seven transgenic lines, with enhanced E $\beta$ f secretion.

Research into two of the cell lines, E $\beta$ f-1 and E $\beta$ f-2, showed that there is a distinct genetic difference between them, and both are tagged with a different genetic marker; E $\beta$ f-1 with genetic marker 1 and E $\beta$ f-2 with genetic marker 2, as shown in Fig. 3.1.

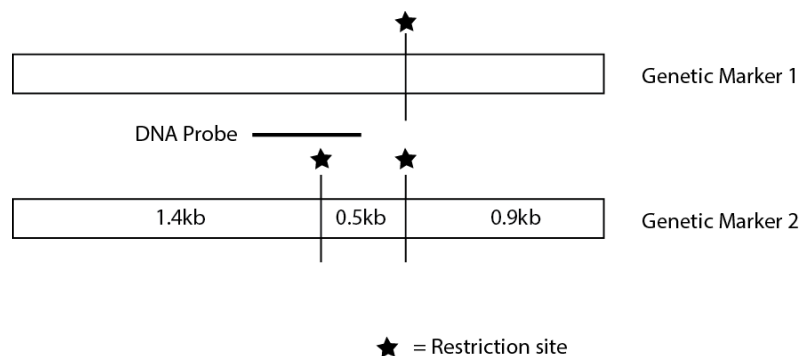


Fig. 3.1

- (b) (i) Explain how restriction fragment length polymorphism comes about. [2]

- Difference in nucleotide sequences between chromosomes;
- causes restriction sites to be created / disappear at a loci;

(ii) With reference to Fig. 3.1, describe how RFLP can be used to identify cell line carrying E $\beta$ -2 gene. [4]

- Difference in DNA sequence of between the two cell lines result in **differences in restriction enzyme recognition site** and result in different DNA fragments;

[Max 1]

- DNA restriction fragments are separated according to **molecular size / molecular weight/ molecular mass** by agarose **gel electrophoresis**;
- **Fragments of lower molecular mass / weight migrate faster**;
- Agarose gel placed in **NaOH** to separate DNA into **single-stranded DNA, transferred onto nitrocellulose membrane** by capillary action;
- Nitrocellulose membrane is incubated with **single-stranded radioactively labelled nucleic acid probe** which will **hybridise to specific target nucleotide sequences** by complementary base pairing;
- **Autoradiography** carried out and X-ray film is exposed to X-ray. Target DNA bands bound to radioactively labeled probes will show up as a **black band** on the X-ray film;

[Max 2]

- By comparing the DNA bands in the DNA/genetic fingerprint, one can detect and observe the difference between E $\beta$ -1 (1.45kb) and E $\beta$ -2 (1.4 kb and 0.5kb) allele;

[Max 1]

[Total: 15]

#### 4 Planning Question

Enzymatic reactions are influenced by various factors, such as the presence of inhibitors. There are two main types of inhibitors, the competitive inhibitors and non-competitive inhibitors. The mode of action of the two types of inhibitors differs and both have a different effect on enzyme-catalysed reactions.

It was known that the action of amylase is inhibited by the presence of ions such as copper ( $\text{Cu}^{2+}$ ). Plan an investigation to determine if  $\text{Cu}^{2+}$  functions as a competitive inhibitor or a non-competitive inhibitor of amylase.

Your planning must be based on the assumption that you have been provided with the following apparatus and materials, as well as other appropriate apparatus and equipment found in the school laboratory.

- 1 mol dm<sup>-3</sup> amylase enzyme
- 5 g dm<sup>-3</sup> starch solution
- Benedict's reagent
- 5 mol dm<sup>-3</sup> Copper Sulphate ( $\text{CuSO}_4$ ) stock solution
- 10 mol dm<sup>-3</sup> maltose solution
- Glass rods
- White tiles
- Bunsen burner, tripod and wire gauze
- Stopwatch
- Access to water bath
- Other measuring instruments (measuring cylinder, syringes, etc.) and apparatus (beakers, test-tube, etc.)

Your plan should have a clear and helpful structure to include:

- Have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it, including scientific reasoning behind procedure.
- Be illustrated by relevant diagram(s) to show, for example, the arrangement of apparatus used.
- Include the layout of result tables and graphs with clear headings and labels.
- Include full details and explanations of the procedures that you would adopt to ensure that the results obtained were as quantitative, precise and reliable as possible.
- Include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

### Suggested answers

#### Background theory: max 3 marks

#### Methodology: max 6 marks

##### ▪ Preparation of maltose colour standards

1. Prepare 5 different concentrations of maltose solutions according to the dilution table shown below. The volume of each concentration will be the same at 10cm<sup>3</sup>.

| Concentration of maltose (mol dm <sup>-3</sup> ) | Volume of stock maltose solution (cm <sup>3</sup> ) | Volume of distilled water (cm <sup>3</sup> ) |
|--------------------------------------------------|-----------------------------------------------------|----------------------------------------------|
| 10.0                                             | 10.0                                                | 0.0                                          |
| 8.0                                              | 8.0                                                 | 2.0                                          |
| 6.0                                              | 6.0                                                 | 4.0                                          |
| 4.0                                              | 4.0                                                 | 6.0                                          |
| 2.0                                              | 2.0                                                 | 8.0                                          |
| 0.0                                              | 0.0                                                 | 10.0                                         |

[1]

2. Prepare a boiling water bath using the beaker and Bunsen burner provided.
3. To 2 cm<sup>3</sup> of 10mol dm<sup>-3</sup> maltose, add 2 cm<sup>3</sup> of benedict's solution. Mix well by stirring with a glass rod. [1]
4. Repeat step 4 for all other maltose concentration.
5. Place all the test-tubes into the boiling water bath for 2 min.
6. After 2 min, remove all the test tubes, note the colour change and place them in a test-tube rack. The tubes can be used as color standards.

##### **OR**

After 2 min, remove the test-tubes. Stir each suspension and ¾ fill a cuvette with each suspension and measure absorbance using a colorimeter. Draw a standard graph. [1]

##### ▪ Preparation of different concentration of starch

1. Prepare 5 different concentrations of starch according to the dilution table below. Make up the total volume to 10cm<sup>3</sup>.

| Concentration of starch (g dm <sup>-3</sup> ) | Volume of stock starch solution (cm <sup>3</sup> ) | Volume of distilled water (cm <sup>3</sup> ) |
|-----------------------------------------------|----------------------------------------------------|----------------------------------------------|
| 5.0                                           | 10.0                                               | 0.0                                          |
| 4.0                                           | 8.0                                                | 2.0                                          |
| 3.0                                           | 6.0                                                | 4.0                                          |
| 2.0                                           | 4.0                                                | 6.0                                          |
| 1.0                                           | 2.0                                                | 8.0                                          |
| 0.0                                           | 0.0                                                | 10.0                                         |

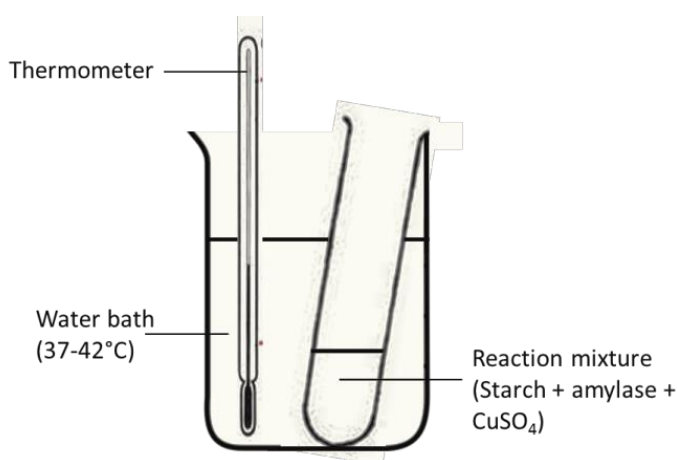
[1]

2. Acclimatize the amylase solutions and starch suspension separately, in a 38-42°C water bath for 2 min. [1]
3. After 2 min, add 1cm<sup>3</sup> of the 5% CuSO<sub>4</sub> solution and 1cm<sup>3</sup> amylase solution to 5cm<sup>3</sup> starch solution.
4. Mix well by stirring and start time. Allow the reaction to take place for 1 minute.
5. Immediately after 1 minute, remove the reaction mixture.
6. To 2cm<sup>3</sup> of the reaction mixture, add equal volume of benedict's solution and carry out Benedict's test.

7. Compare the colour change with the colour standard /standard graph prepared earlier to determine the concentration of maltose produced. [1]
8. Repeat steps 2 to 9 using fresh solutions to obtain two further readings.
9. Repeat steps 2 to 10 using each concentration of starch solution.
10. Repeat steps 2 to 11 / repeat the entire experiment, but do not add  $\text{CuSO}_4$  solution this time. This is to obtain results of the effect of [starch] on the rate of starch hydrolysis / enzyme activity on starch hydrolysis without the presence of inhibitor. [1]
11. **Control:** Replace amylase with distilled water.

**Purpose:** To show that any production of glucose still due to the presence of amylase which hydrolyses starch, and hence the results obtained with and without  $\text{CuSO}_4$  is due to the  $\text{CuSO}_4$  effect on amylase. [1]

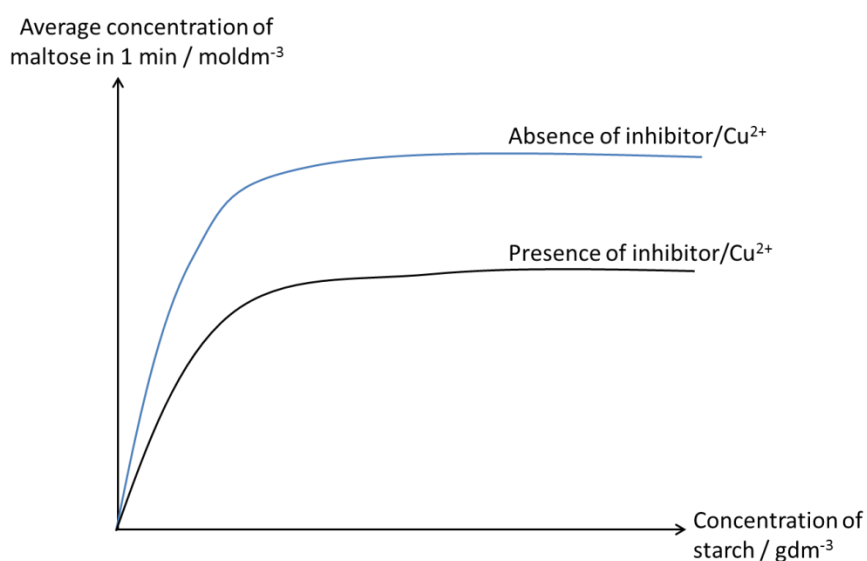
**Diagram [1]**



**Results – Table [1 for correct proposed layout]**

| Concentration of starch / $\text{gdm}^{-3}$ | Concentration of maltose concentration produced in the <b>presence</b> of $\text{Cu}^{2+}$ in 1 min/ $\text{mol dm}^{-3}$ |    |    |         | Concentration of maltose concentration produced in the <b>absence</b> of $\text{Cu}^{2+}$ in 1 min/ $\text{mol dm}^{-3}$ |    |    |         |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----|----|---------|--------------------------------------------------------------------------------------------------------------------------|----|----|---------|
|                                             | R1                                                                                                                        | R2 | R3 | Average | R1                                                                                                                       | R2 | R3 | Average |
| 5.0                                         |                                                                                                                           |    |    |         |                                                                                                                          |    |    |         |
| 4.0                                         |                                                                                                                           |    |    |         |                                                                                                                          |    |    |         |
| 3.0                                         |                                                                                                                           |    |    |         |                                                                                                                          |    |    |         |
| 2.0                                         |                                                                                                                           |    |    |         |                                                                                                                          |    |    |         |
| 1.0                                         |                                                                                                                           |    |    |         |                                                                                                                          |    |    |         |
| 0.0                                         |                                                                                                                           |    |    |         |                                                                                                                          |    |    |         |

### **Results – Graph [1 for correct proposed graphs]**



Copper ion functions as a non-competitive inhibitor by binding to a site away from the active site. The binding alters the tertiary structure of the enzyme and prevents the rate of reaction from reaching  $V_{\text{max}}$ .

### **Risk assessment [1]**

| <b>Risk</b>                                                          | <b>Precaution</b>                                                                                                                |
|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Copper sulfate / amylase /starch solution is an irritant to the skin | Wear gloves and goggles when handling / Wash hands thoroughly if hands come into contact with solution.                          |
| Boiling water may scald / burn skin                                  | Use a test-tube holder to hold test-tube when transferring test tube from boiling water bath after carrying out Benedict's test. |

## Free response question

Write your answers to this question on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate,
- must be in continuous prose, where appropriate,
- must be set in sections **(a)**, **(b)** etc., as indicated in the question.

- 5 (a) Describe cystic fibrosis and how it can be treated using a non-viral gene [8]  
delivery system.

### Cystic fibrosis description (max 4)

1. Cystic fibrosis (CF) is a autosomal recessive disease;
2. caused by mutation of cystic fibrosis transmembrane conductance regulator\* (CFTR/ CFTCR) gene;
3. The mutation of CFTR gene in chromosome 7 is a deletion of 3 bases resulting in the loss of phenylalanine in the polypeptide;
4. A normal CFTR gene will code for a protein that function as a gated channel protein to facilitate the transport of chloride ions;
5. as a result of this loss of function\* mutation, channels no longer able to transport chloride ions out of the epithelial cell (as in the lung);
6. causing thick mucus in lungs, resulting in blockage/congestion of airway in the lung that reduces gaseous exchange leading to breathing difficulties

Or

pancreatic duct is choked by thick mucus preventing release of enzymes resulting in indigestion ;

7. thick mucus provides suitable environment for bacteria growth which makes patient susceptible to chronic lung infection resulting in death (lowering of life expectancy, not living beyond 30 years old);

### Treatment (max 4)

8. to treat CF, a normal functioning allele\* of CFTR is isolated and inserted into a plasmid vector through recombinant techniques;
9. this DNA construct is then packaged in the aqueous core of the microscopic lipid envelopes/phospholipid bilayer sphere forming liposomes\* ;
10. delivered to epithelial cell in the lung by aerosol spray (nasal spray);
11. the liposome fuses with the plasma membrane of the epithelial cell lining the lung and releases the DNA into the cell (this process is called lipofection);
12. the plasmid vector enters nucleus and the normal CFTR allele is transcribed and translated;
13. to produce normal CFTR protein which can be incorporated into plasma membrane to facilitate the transport of chloride;

(b) Describe how the micropropagation technique is used in cloning and production of herbicide resistant plants. [7]

1. Obtain **mRNA for glyphosate-resistant EPSPS gene** and using reverse transcription method, produce the **double-stranded cDNA** (*or any sound method to isolate this gene*);
2. Add linker DNA to both ends of the gene of interest; Remove the oncogenes and opine producing genes from the **Ti plasmids** of *Agrobacterium*;
3. cut the Ti plasmid using the same restriction enzyme as that for Bt gene, forming complementary sticky ends and mix the cut plasmid and gene of interest/ to allow complementary base-pairing via hydrogen bonds to temporarily hold the fragments together between plasmid and gene of interest after using same restriction enzymes;
4. Add DNA ligase to seal the nicks in the sugar-phosphate backbone forms phosphodiester linkages between fragments;
5. Insert recombinant Ti-plasmids back into the Agrobacterium;
6. In micropropagation, plant tissues or explants e.g.: meristematic cells are removed from plants (*Reject: explants only*);
7. **Infect the explants with Agrobacterium containing the recombinant Ti plasmids. Select explants** that were successfully transformed;
8. Surface of explants sterilised with dilute sodium hypochlorite (Clorox) to kill bacterial and fungal pathogens or organisms;
9. and grown in a containing sterile media containing nutrients and plant growth regulators needed for plant growth; /Containers holding the explants are then sealed and incubated for 1-9 weeks.
10. During this period, the cultured cells divide by mitosis to form a **mass of undifferentiated tissue** called a callus;
11. By adjusting concentration of **auxin:cytokinin ratio** in growth medium; cells in callus can be induced to **differentiate into shoots followed by roots**;
12. Further growth being encouraged by the use of plant growth substances until plantlets are large enough;/To be weaned and planted in sterile soil in green house and acclimatization in green house, the plant are transferred to soil for field planting;

(c) Discuss the social and ethical implications of genetically modified plants using a named example. [5]

1. Definition of GM plant – 1m

**Genetically Modified plant:** Plant that has acquired one or more genes by artificial means. The genes may or may not be from the same species.

- 1 named example + description - 1m

2. Social implications [max 2m]

(a) Threat to Human Safety

**Genetic markers**

- Vectors used for transforming plant /crop cells – contain antibiotic markers/ kanamycin resistance;

- Concerns that the antibiotic resistance gene if is not properly broken down by the digestive system upon consumption of plant, may be passed to potentially harmful bacteria in the human gut making them resistant to the antibiotics;

**(b) Threat to Safety of the Environment**

**(i) Genetically modified crops might establish themselves as weeds (Bt corn, Roundup Ready soybean, any valid e.g.)**

- Seeds from the transgenic crops that have been modified to withstand unfavourable environmental conditions may be carried by the wind to other places. These crops grow quickly and may establish themselves as weeds;

**(ii) Spread of resistance from genetically modified crops to weeds (Any herbicide resistant plant)**

- Pollen grains from genetically modified plants can be carried in the wind and fertilise wild relatives and weeds;
- leading to formation of “super weeds”. These may become more invasive (overrun agricultural areas rapidly) or compete with the natural species;

**(iii) Upsetting the ecological balance (Bt corn etc.)**

- the genetically modified plants themselves may out-compete other plants in the same area;
- → this may upset the balance of the native plant species / decrease in biodiversity;

**3. Ethical implications [max 2m]**

**(a) Genetic manipulation of plants may be deemed unacceptable as there are no rights to tampering with nature and affecting the intrinsic value of the plant;**

**(b) Is it acceptable to engineer any plant to be used as food for human consumption? (Give e.g.)**

- Plants are not biologically ‘designed’ to withstand the stress of producing additional proteins / toxins;

**Golden rice / Bt Corn**

**(c) Genetic engineering to make a staple crop more resistant in marginal conditions could be a danger of increased dependence on rich “developed” countries**

- i.e. richer countries benefiting at the expense of the poorer countries;
- world food production may be dominated by a small number of large companies with the technical know-how;

**Other points:**

**(d) Non-mandatory labelling of products containing GMO plants → consumer rights;**

**(e) Patent rights to GMO plants → patenting plants is itself unethical as it reduces them to the level of objects;**

**[Total: max 5]**

[Total: 20]

**End of paper**